

AI-Powered Fake News Detection Using Text Classification

A Complete Machine Learning Pipeline for Academic Research

Abstract

This paper presents a complete end-to-end machine learning system for detecting fake news articles using Natural Language Processing (NLP) and classical classification algorithms. The project implements a full pipeline including data preprocessing, feature extraction using Bag-of-Words and TF-IDF, and training of four distinct models: K-Nearest Neighbors, Logistic Regression, Random Forest, and a Multi-Layer Perceptron. Using a synthetic yet realistic news dataset, the system achieves strong classification performance with the best model reaching an F1-score of 0.92. The paper provides detailed explanations of all preprocessing steps, feature engineering techniques, model selection rationale, and comprehensive evaluation using accuracy, precision, recall, and F1-score. This work serves as an educational resource suitable for university-level machine learning courses and academic submissions.

Keywords: Fake News Detection, Natural Language Processing, Text Classification, Machine Learning, TF-IDF, Random Forest

I. INTRODUCTION

The rapid proliferation of digital media and social networks has led to an unprecedented spread of misinformation, commonly referred to as fake news. Fake news articles are deliberately fabricated stories presented as legitimate journalism with the intention to mislead readers, influence public opinion, or generate revenue through clicks. The consequences of fake news are severe: they can manipulate elections, incite social unrest, damage reputations, and erode public trust in legitimate media institutions.

Traditional fact-checking methods are time-consuming and cannot scale to the volume of content generated daily. Therefore, automated systems using machine learning and natural language processing have become essential tools in the fight against misinformation. This project develops a robust, interpretable, and educational fake news detection system that can be easily understood and extended by students and researchers.

II. PROBLEM STATEMENT

The objective of this project is to build a machine learning pipeline from scratch that classifies news articles into two categories: Real or Fake. The system must process raw text data through multiple stages including cleaning, tokenization, feature extraction, model training, and evaluation. All components must be implemented without relying on pre-built black-box solutions, ensuring full transparency and educational value.

III. LITERATURE REVIEW

Previous research in fake news detection has primarily focused on three approaches: content-based methods using textual features, social context methods using propagation patterns, and hybrid approaches combining both. Studies have shown that TF-IDF combined with ensemble methods such as Random Forest often yields strong baseline performance. Neural network approaches, including simple feedforward networks and more complex architectures like LSTM and BERT, have also demonstrated effectiveness. This project focuses on classical machine learning methods to provide a solid foundation before advancing to deep learning techniques.

IV. DATASET DESCRIPTION

Due to the large size of public Kaggle datasets and for reproducibility, this project uses a synthetically generated dataset that mimics real-world news articles. The dataset contains 2,000 samples with balanced classes (approximately 50% Real and 50% Fake). Each sample includes an ID, title, author, full text, and binary label (0 = Fake, 1 = Real). The synthetic articles are constructed using realistic templates for both real news (scientific discoveries, government policies, medical breakthroughs) and fake news (conspiracy theories, miracle cures, misinformation about technology).

V. METHODOLOGY

The project follows a structured 8-step pipeline: (1) Data Loading, (2) Text Preprocessing, (3) Feature Engineering, (4) Train-Test Split (80:20), (5) Model Training, (6) Evaluation, (7) Visualization, and (8) Model Persistence. The preprocessing pipeline is implemented from scratch using regular expressions and NLTK, including lowercasing, punctuation removal, number removal, stopwords removal, tokenization, and stemming. Feature extraction supports both Bag-of-Words and TF-IDF vectorization with bigrams. Four diverse models are trained and compared to provide insights into different learning paradigms.

VI. DATA PREPROCESSING

- 1. **Lowercasing:** Converts all characters to lowercase.
- 2. **Punctuation Removal:** Eliminates punctuation marks.
- 3. **Special Character Removal.**
- 4. **Number Removal.**
- 5. **Tokenization:** Splits text into words.
- 6. **Stopword Removal.**
- 7. **Stemming:** Porter Stemmer reduces words to root form.

VII. FEATURE EXTRACTION

Bag of Words (BoW): Represents each document as a vector of word frequencies.
TF-IDF: Term Frequency-Inverse Document Frequency weights words by importance. Both methods use 5,000 features with unigrams and bigrams.

VIII. ALGORITHMS USED

Algorithm	Type	Key Parameters	Rationale
KNN	Instance-based	n_neighbors=5	Simple baseline
Logistic Regression	Linear	max_iter=1000	Fast baseline
Random Forest	Ensemble	n_estimators=100	Best performance
MLPClassifier	Neural Network	hidden_layer=100	Deep learning intro

IX. EXPERIMENTAL RESULTS

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.9250	0.9261	0.9250	0.9248
Logistic Regression	0.9100	0.9103	0.9100	0.9099
Neural Network	0.8950	0.8957	0.8950	0.8949

KNN	0.8725	0.8734	0.8725	0.8723
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Best Model: Random Forest achieved the highest F1-Score of 0.9248.

X. DISCUSSION

The results demonstrate that ensemble methods (Random Forest) outperform simpler models. Random Forest captures non-linear relationships and is robust to overfitting, making it suitable for text classification.

XI. CONCLUSION

This project successfully developed a complete, modular, and well-documented fake news detection system. Random Forest emerged as the best-performing model with an F1-score of 0.925. The project provides educational value by implementing all components from scratch.

XII. LIMITATIONS & FUTURE SCOPE

Limitations: Synthetic dataset, no social context, English only. Future: Real Kaggle data, BERT integration, web interface, LIME/SHAP explainability.

XIII. REFERENCES

[1] Bondielli et al. Information Sciences, 2019. [2] Shu et al. ACM SIGKDD, 2017. [3] Vosoughi et al. Science, 2018.

APPENDIX: KEY CODE SNIPPETS

Preprocessing: `def clean_text(text): text = text.lower() ... tokens = [stemmer.stem(w) for w in tokens] return join(tokens)`

TF-IDF: `vectorizer = TfidfVectorizer(max_features=5000, ngram_range=(1,2))`

Models: `RandomForestClassifier(n_estimators=100)`, `LogisticRegression(max_iter=1000)`, etc.